The Complete Ordered Field Is Unique

Chapter 5 ended with one complete ordered field, whose elements are classes of Cauchy sequences of rationals. *Axiomatic Set Theory* builds another, whose elements are sets of rationals, and *Analysis on the Real Line*, Chapter 1 builds none: it assumes that a complete ordered field exists and calls it $\mathbb{R}$. Are these the same $\mathbb{R}$? The axioms of an ordered field alone do not decide it. The field $\mathbb{Q}(t)$ of rational functions of Chapter 4, ordered by leading coefficients, satisfies all of them and has an element above every natural number; the field $\mathbb{Q}(\sqrt{2})$ is even Archimedean, and still 3 has no square root in it. This chapter proves that completeness is the property that pins $\mathbb{R}$ down: any two complete ordered fields are linked by an order-preserving field isomorphism, and by only one. So "the" real numbers are well defined, and no theorem about them depends on how they were built.

The tool is one idea. In an Archimedean ordered field an element is determined by **the rationals below it**, and a map that respects sums, products, 1 and the order must send each element to one with the same rationals below it. In a complete field the supremum of the copies of those rationals is such an element. So from an Archimedean field into a complete one there is exactly one order-preserving field map, and each identity it satisfies is checked by comparing rationals below the two sides, with no arithmetic of suprema. When both fields are complete, maps run both ways, and the word "exactly" makes them inverse to each other without a computation. The same uniqueness leaves a complete ordered field no symmetry: a map of it to itself that respects only sums, products and 1 is the identity, because there the positive elements are the non-zero squares. Without the order, a field like $\mathbb{Q}(\sqrt{2})$ does have a symmetry that breaks it; that example belongs to *Algebraic Preliminaries*, Chapter 2, and this chapter only points to it. The chapter, the last of the volume's main line, ends by comparing the two ways to begin analysis, assuming the axioms or building a model, and by listing what *Analysis on the Real Line* takes from this volume.

What you need. Chapters 4 and 5 above all, with Chapters 2 and 3; *Sets and Functions*, Chapters 1, 2, 4 and 5, chiefly Section 5.3; and *Logic and Proof*, Chapters 2 and 6, with Chapters 5 and 7 for one remark. From Chapter 4 of this book: the rules of an ordered field and the absolute value (Proposition 4.1.2, Lemma 4.1.1, Definition 4.1.1, Proposition 4.1.3); powers and the one copy ι_F of $\mathbb{Q}$ in an ordered field, with its uniqueness clause (Definition 4.2.1, Theorem 4.2.2); the Archimedean property, density of $\mathbb{Q}$, and the non-Archimedean field $\mathbb{Q}(t)$ (Definition 4.2.2, Proposition 4.2.3, Theorem 4.2.5, Example (A Non-Archimedean Ordered Field)); and, in Section 6.3 and its exercises, limits and Cauchy sequences (Definition 4.3.2, Example (First Limits), Theorem 4.4.3, Proposition 4.4.4, Definition 4.5.1, Proposition 4.5.1, Theorem 4.6.1). From Chapter 5: complete ordered fields (Definition 5.5.1); the Cauchy model $\mathbb{R}$, its classes and its order (Definition 5.1.1, Definition 5.1.2, Definition 5.2.1, Theorem 5.3.3, Lemma 5.3.4); the theorems that it is a complete ordered field (Theorem 5.5.2) and Archimedean (Theorem 5.4.2); the map $\kappa = \iota_{\mathbb{R}}$ (Definition 5.4.1, Theorem 5.4.1); and the class of the Newton sequence, a positive real number whose square is 2 (Example (The Class of the Newton Sequence)). From Chapter 3: $\mathbb{Q}$ as classes of pairs (Definition 3.1.1, Proposition 3.1.1), $\mathbb{Q}$ Archimedean (Theorem 3.2.3), reciprocals and the order (Proposition 3.2.2), no rational square equal to 2 (Theorem 3.3.1, Theorem 3.3.2), and the countability of $\mathbb{Q}$ (review exercise R2 of Section 3.3). From Chapter 2: the axioms (F1)–(F9) and (O1)–(O4) (Definition 2.3.1, Definition 2.4.1) and their rules (Proposition 2.3.2, Proposition 2.4.3). Chapter 1 supplies one analogy (Theorem 1.4.1) and, in Exercise C3 of Section 6.3, induction and the order arithmetic of $\mathbb{N}$ (Theorem 1.1.2, Proposition 1.3.1). From *Sets and Functions*: proving uniqueness, Section

1.4 (Sets and Functions, Example (Method: existence and uniqueness)); injective and bijective maps, compositions and two-sided inverses, Sections 2.2 and 2.4 (Sets and Functions, Definition 2.2.1, Sets and Functions, Definition 2.2.3, Sets and Functions, Theorem 2.2.1, Sets and Functions, Theorem 2.4.4, Sets and Functions, Example (Method: construct and verify)); countable sets, Section 4.2, for exercises of Section 6.3 (Sets and Functions, Theorem 4.2.2, Sets and Functions, Proposition 4.2.4); equivalence relations and well-definedness, Section 5.2 (Sets and Functions, Definition 5.2.1, Sets and Functions, Example (Method: checking that a definition on classes is well defined)); and upper bounds and suprema, Section 5.3 (Sets and Functions, Definition 5.3.2, Sets and Functions, Definition 5.3.5). From *Logic and Proof*: the contrapositive, Chapter 2 (Logic and Proof, Proposition 2.3.2); the negation of quantifiers, unbounded and bounded, Chapter 6 (Logic and Proof, Proposition 6.2.1, Logic and Proof, Proposition 6.2.3); and, for the remark on what an isomorphism carries in Section 6.3, first-order formulas and their invariance under isomorphism, Chapters 5 and 7 (Logic and Proof, Definition 5.2.2, Logic and Proof, Theorem 7.5.1).

Time. About 15 hours on the core track (all of A, one B and one C per section, the exit checks); about 25 hours for everything.

6.-1.1 Roadmap

- **What Completeness Gives at Once.** Order-preserving field maps, which reflect the order and fix the rationals; the theorem that a complete ordered field is Archimedean, by a supremum of the copy of $\mathbb{N}$ that cannot exist; the rationals below an element, $\mathbb{Q}_F(x)$, which determine x when F is Archimedean (rationals separate); and square roots in a complete ordered field, built as a supremum and unique.
- **Building the Map from the Rationals Below.** The rationals below a sum, and below a product of positive elements, computed from those below each term; and the main theorem in its one-sided form: from an Archimedean ordered field F into a complete one L there is exactly one order-preserving field map, which sends x to the supremum in L of the copies of the rationals below x.
- **Two Complete Ordered Fields, One Isomorphism.** Any two complete ordered fields are isomorphic by exactly one order-preserving field isomorphism, with the inverse obtained from the uniqueness clause instead of computed; every complete ordered field a copy of the Cauchy model, and a Dedekind cut read as a set of rationals below; the automorphisms of a complete ordered field, of which there is one, the identity, even among the maps that respect only sums, products and 1; what an isomorphism carries (limits, as the worked case) and what it does not (a model whose zero is the empty set); and the two ways to begin analysis, with a table of what *Analysis on the Real Line*, Chapter 1 assumes and where this series supplies it. The review exercises return to Chapter 5's real numbers, to well-definedness on $\mathbb{Q}$ as a set of classes of pairs, and to countability and the density of the elements of a complete ordered field that are not rational.

6.-1.2 Named moves

- **Rationals separate** (Lemma 6.1.3). To show $a = b$ in an Archimedean ordered field, show that the same rationals lie below each; otherwise a rational would lie strictly between them. It is proved in Section 6.1 and then proves every equation between elements of the complete field in the main theorem of Section 6.2 (Steps 3, 5, 6 and 7); the exit check C1 of Section 6.2 runs it alone. In $\mathbb{Q}(t)$, where Archimedes fails, t and $t+1$ have the same rationals below them (the fold in Section 6.1).
- **Construct a point as a supremum** (Example (Method: construct a point as a supremum)). To produce an element you cannot name, take the supremum of a well-chosen set

and refute the two wrong cases of trichotomy, one with an element of the set above the supremum, the other with a smaller upper bound. It builds square roots in Section 6.1 and the map of Section 6.2. Exercise B2 of Section 6.2 (cube roots) leaves the last step to you, and Exercise C3 of Section 6.3 (a bounded increasing sequence converges to the supremum of its terms) the last half.

- **Uniqueness gives inverses** (Sets and Functions, Example (Method: existence and uniqueness)). If the only order-preserving field map $F \to F$ is the identity, then two maps $F \to L \to F$ compose to the identity without computing anything. It proves in Section 6.3 that the map between two complete ordered fields is bijective, and the same uniqueness clause shows there that a complete ordered field has no order-preserving automorphism except the identity; once square roots recover the order, also no field automorphism.

6.-1.3 Threads

- **Rationals below an element.** The copy ι_F of $\mathbb{Q}$ in an ordered field and density (Chapter 4) are what let Section 6.1 define $\mathbb{Q}_F(x)$; Section 6.2 turns it into a map, and Section 6.3 into the uniqueness theorem.
- **Square roots.** Chapter 3 shows that 2 has no square root in $\mathbb{Q}$, Chapter 5 builds one in $\mathbb{R}$ (Example (The Class of the Newton Sequence)), and Section 6.1 proves that every positive element of a complete ordered field has one; Section 6.3 uses them to recover the order from the arithmetic.
- **Categoricity.** Section 1.4 proved that any two Peano systems are isomorphic by exactly one isomorphism; Section 6.3 proves the same for complete ordered fields, with the same shape: existence by a recursion or a supremum, uniqueness by induction or by rationals separate.
- **Two ways to begin analysis.** Section 1.5 says where $\mathbb{N}$ comes from, Chapter 5 builds $\mathbb{R}$ on it, and Section 6.3 compares that route with assuming the axioms, as *Analysis on the Real Line* does.

6.-1.4 Summary

An order-preserving field map between ordered fields fixes the rationals and reflects the order (Definition 6.1.1, Proposition 6.1.1). A complete ordered field is Archimedean (Theorem 6.1.2), and in an Archimedean ordered field an element is determined by the set of rationals below it (Definition 6.1.2, Lemma 6.1.3). Every positive element of a complete ordered field has a square root (Proposition 6.1.4). From an Archimedean ordered field into a complete one there is exactly one order-preserving field map, which sends x to the supremum of the copies of the rationals below x (Theorem 6.2.3). Hence any two complete ordered fields are isomorphic by exactly one order-preserving field isomorphism (Theorem 6.3.1), every complete ordered field is a copy of the Cauchy model (Corollary 6.3.2), and a complete ordered field has no order-preserving automorphism but the identity (Corollary 6.3.3), nor any field automorphism (Theorem 6.3.4).

What is open: $\mathbb{R}$ is now one structure, and nothing here says anything about its sequences, series or continuous functions. *Analysis on the Real Line* starts there, from the axioms this volume has shown to describe exactly one thing.

6.-1.5 Terms

- **order-preserving field map**: a map $\chi\colon F \to L$ of ordered fields that respects sums, products and 1 and is strictly increasing (Definition 6.1.1).
- **rationals below an element**: $\mathbb{Q}_F(x) = \{q \in \mathbb{Q} : \iota_F(q) < x\}$ in an ordered field F (Definition 6.1.2).

- **complete ordered field**: an ordered field in which every non-empty set bounded above has a supremum (Definition 5.5.1).
- **isomorphic complete ordered fields**: linked by an order-preserving field isomorphism, which is unique (Theorem 6.3.1).